Fig 1. Basidium paper model

Fig 2. Basidiomycota paper model

I decided to create a 3-dimensional Basidiomycota model made out of crepe paper, including a separate enlarged model of a basidium, with labels indicating the different parts of the mushroom. Oftentimes, key reproductive systems such as the basidium and basidiospores are microscopic and hard to observe. Hence, most people are unaware that fungi possess such complex structures that play crucial roles in their reproduction. By enlarging these structures, the model allows viewers to better understand how mushrooms produce and release their spores. I also decided to fashion the gills in a honeycomb lattice pattern to simulate a porous structure. While many people are familiar with the presence of gills in mushrooms, fewer are aware that some fungi, such as Boletus, can have a porous membrane instead of gills. This design choice highlights the diversity of reproductive surfaces found among Basidiomycota. Learning about these structures can help people understand how fungi have evolved various adaptations to produce and disperse spores efficiently, influencing the way that these fungi spread and colonise new habitats.

Furthermore, creating a 3- dimensional model allows viewers to better visualise fungal anatomy compared to 2- dimensional diagrams. The spatial arrangement of the different structures of the mushroom become much easier to understand when represented physically. Labelling the parts of a mushroom also helps improve public awareness of anatomical features, supporting mushroom identification and raising awareness of the different toxic and edible species.

Finally, I believe that this model is also able to invoke a sense of curiosity and wonder. By learning about the anatomical structure of a mushroom, viewers may become more attentive when encountering fungi. With a better understanding of mushroom anatomy, viewers may begin to observe characteristic features such as gills, pores or other structures, allowing them to gain a deeper appreciation for the diversity and complexity of fungi.